

## Highlights

### **A Comparative Study of Emerging Architectures for EDA-based Arousal Classification**

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- Mamba achieves  $F1 = 0.858$  at 1.5 ms, approaching PatchTST with  $3.6\times$  lower latency
- Eight architectures spanning five paradigms compared under strict LOSO validation
- Accuracy-efficiency Pareto frontier with five complementary computational metrics
- State space models match attention-based methods on physiological time-series
- Derivative channels most beneficial for simpler architectures; attention learns similar context

# A Comparative Study of Emerging Architectures for EDA-based Arousal Classification

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## Abstract

Electrodermal Activity (EDA) is a non-invasive biomarker of sympathetic arousal increasingly used in wearable affective computing. While Transformer-based architectures outperform convolutional baselines for subject-independent arousal classification, their computational cost limits deployment on latency-sensitive platforms. This work presents a systematic comparison of eight emerging architectures spanning five paradigms—patch-based attention, sparse attention, frequency-domain modelling, state space models, and modernised convolutions—for binary arousal classification from EDA signals under strict Leave-One-Subject-Out (LOSO) validation with 147 participants. Beyond predictive accuracy, we introduce five complementary computational efficiency metrics—parameter count, FLOPs, inference time, memory footprint, and training time—as primary evaluation dimensions, enabling Pareto-frontier characterisation. Mamba, a selective state space model, achieves  $F1 = 0.858$

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at 1.5 ms inference time, approaching PatchTST accuracy ( $F1 = 0.863$ ; difference does not survive Bonferroni-Holm correction) with  $3.6\times$  lower latency and 66% fewer parameters. Channel ablation further suggests derivative channels provide greatest benefit for simpler architectures, while the temporal context from derivatives is partially redundant with representations learned by attention and state-space mechanisms. A classical signal processing baseline achieved  $F1 \approx 0.81$ , demonstrating a  $\sim 5\%$  improvement from architectures with global temporal modelling over both classical and simple learned baselines. These results offer practical guidance for architecture selection under computational constraints. Although these findings are based on a single 147-participant laboratory dataset, consistency across LOSO folds supports reliability; cross-dataset validation is needed to confirm generalisability.

*Keywords:* Electrodermal Activity, Arousal Classification, Efficient Architectures, State Space Models, Affective Computing, Edge Deployment

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## 1. Introduction

Electrodermal Activity (EDA) reflects sympathetic nervous system activation through sudomotor nerve activity. Unlike heart rate or respiration, sweat glands receive no parasympathetic innervation, making EDA a biomarker of exclusively sympathetic arousal [1, 2, 3, 4]. The signal comprises a slow-varying tonic level and rapid phasic skin conductance responses (SCRs) reflecting immediate sympathetic activation [5]. The growing availability of wearable EDA sensors [6] has spurred interest in deploying automatic arousal classifiers on edge devices for mental health monitoring, stress detection, and adaptive human-computer interaction [7, 8]. However, practical deployment requires architectures that simultaneously achieve high subject-independent accuracy, low computational cost, and minimal memory—a combination that has received limited systematic investigation.

Several deep learning approaches have been proposed for EDA-based affect recognition. Convolutional networks operating on time-frequency EDA representations have shown promise for emotion classification [9], while 1D-CNNs processing raw SCR waveforms achieved subject-independent arousal detection [10]. Long Short-Term Memory (LSTM) and bidirectional LSTM networks, widely adopted for physiological time-series classification [11], were excluded from this benchmark. Recurrent architectures were found to un-

derperform convolutional and attention-based models under LOSO in companion experiments on the same dataset, likely because the relatively short 40-second windows limit the benefit of explicit memory gating relative to direct temporal convolution or attention [12]. Transformer-based methods have been applied to EDA decomposition [13] and multimodal physiological recognition [14].

Within BSPC, joint analysis of multidomain skin conductance features has been applied to psychological stress identification [15], wrist-worn EDA sensors have enabled autonomic sleep staging [16], and multimodal physiological approaches have been proposed for emotion recognition [17]. Recent BSPC contributions further include CNN-Transformer hybrid architectures for stress detection from photoplethysmography signals [18] and explainable deep neural networks combining biosignals with human-engineered features for stress classification [19]. The broader literature includes systematic architecture comparisons for EEG [11] and ECG [20, 21], but a methodologically rigorous comparison under strict Leave-One-Subject-Out (LOSO)—which prevents the subject-level data leakage known to inflate performance in physiological signal classification [22]—remains absent for EDA-based arousal classification.

In prior work [12], we compared five architectures (1D-CNN, TCN, InceptionTime, TST, PatchTST) under LOSO with 147 participants. Transformer-based models consistently outperformed convolutional baselines, with PatchTST achieving the best performance ( $F1 = 0.852$ ,  $AUC = 0.912$ ) and attention mechanisms concentrating on physiologically meaningful SCR onset and rising phases. However, despite patch-based tokenisation reducing the effective sequence length, PatchTST retains  $O(N^2)$  self-attention complexity, and its latency and memory consumption on a 40-second EDA window ( $T = 160$  at 4 Hz) may exceed the constraints of low-power wearable processors. This practical limitation motivates investigation of substantially more efficient architectures.

The time-series community has developed efficient Transformer variants with sub-quadratic or linear complexity [23, 24]: Informer uses ProbSparse attention [25], Autoformer replaces attention with FFT-based Auto-Correlation and progressive decomposition [26], FEDformer operates via Fourier-enhanced attention on randomly selected frequency modes [27], and TimesNet transforms 1D series into 2D tensors via FFT-discovered periods for Inception-based 2D convolution [28]. DLinear demonstrated that simple trend-seasonal decomposition with linear layers can outperform sophisticated Transform-

ers on forecasting benchmarks [29, 11]. Beyond Transformers, Mamba introduced selective state space models (SSMs) where parameters are input-dependent, achieving competitive performance on language and genomics [30] and showing promise for physiological signals via bidirectional architectures [20]. ModernTCN extended the convolutional paradigm with large depthwise kernels and inverted bottlenecks, matching Transformer performance while maintaining convolutional efficiency [31, 32].

Despite these advances, no systematic evaluation spanning these paradigms exists for physiological signal classification under rigorous subject-independent protocols. This gap is significant because EDA signals exhibit inter-subject variability, non-stationarity, and structured phasic responses. These signal characteristics may interact differently with each efficiency mechanism compared to the forecasting benchmarks on which these architectures were originally developed [23]. Furthermore, the SSM and modernised convolution paradigms have not been benchmarked against Transformer-based methods on physiological classification, leaving practitioners without evidence to guide paradigm selection.

This work addresses this gap through a systematic equal-budget comparison of eight architectures spanning five paradigms and a linear baseline: PatchTST (patch-based attention), Informer (sparse attention), Autoformer, TimesNet, and FEDformer (frequency-domain), Mamba (selective state space model), ModernTCN (modernised convolution), and DLinear as a linear baseline. All architectures are evaluated on the identical 147-participant dataset and LOSO protocol. Computational efficiency—parameter count, FLOPs, inference time, memory, and training time—is elevated to a primary evaluation dimension alongside classification metrics. The contributions are: (i) a systematic accuracy-efficiency comparison of Transformers, SSMs, and modernised convolutions for EDA under LOSO; (ii) explicit characterisation of the Pareto frontier, directly informing architecture selection under deployment constraints; (iii) inclusion of DLinear and ModernTCN as critical baselines testing whether complex temporal mechanisms provide measurable benefit for EDA; and (iv) physiologically-grounded insights into how each efficiency paradigm interacts with phasic EDA dynamics.

## 2. Method

### 2.1. Dataset and Experimental Protocol

The experimental dataset follows a controlled stress elicitation protocol described in previous work [33]. Electrodermal Activity signals were recorded from 147 healthy participants (aged 18–44 years) exposed to audiovisual stimuli designed to induce calm and stress states under controlled laboratory conditions. Each stimulus lasted 47 seconds. To eliminate transition artefacts associated with stimulus onset and offset, the first 4 seconds and the last 3 seconds were removed, resulting in 40-second effective segments per trial. Signals were acquired at a constant sampling frequency of 4 Hz.

Each participant completed multiple trials in both calm and stress conditions, producing a balanced binary classification dataset. All trials belonging to the same participant were preserved within the same fold during cross-validation to prevent subject-level data leakage—a critical methodological safeguard given the strong inter-subject variability characteristic of EDA signals [1, 22].

### 2.2. EDA Signal Processing and Input Representation

Raw skin conductance signals were preprocessed following the pipeline established in our prior work [33]. A low-pass FIR filter (cut-off frequency: 3.5 Hz, stopband: 4 Hz) was applied to prevent aliasing, followed by Gaussian smoothing to attenuate acquisition noise. Continuous Decomposition Analysis (CDA), a non-negative deconvolution method, was used to decompose the signal into tonic skin conductance level (*SCL*) and phasic skin conductance response (*SCR*) [4, 34, 35]:

$$SC(t) = SCL(t) + SCR(t) \quad (1)$$

Only the phasic component  $SCR(t)$  was retained for modelling, as it directly reflects sympathetic nervous system activation uncontaminated by slow tonic drift. Alternative EDA decomposition approaches include *cvxEDA*, which formulates decomposition as a convex optimisation problem with physiologically grounded constraints on non-negative SCRs and sparse phasic drivers [36], and *Ledalab*, which employs non-negative deconvolution with an a priori SCR shape [35]. CDA was retained from our prior pipeline for consistency; however, the choice of decomposition method is known to influence downstream signal characteristics and classification performance across

different EDA preprocessing pipelines [37]. While the controlled laboratory conditions of the present dataset minimise motion artefacts, EDA signals in ambulatory and real-world settings are susceptible to electrode contact noise and movement artefacts [38]—a factor relevant to the deployment scenarios discussed in Section 3.3. To enrich the short-term dynamic representation and facilitate early detection of sympathetic responses, two derivative channels were computed from the phasic signal: the first temporal derivative ( $\Delta SCR$ ) and the second temporal derivative ( $\Delta^2 SCR$ ). These channels encode the velocity and acceleration of sympathetic activation respectively, providing the model with explicit information about the rate and pattern of phasic change [10].

Each temporal window of length  $n$  seconds was thus represented as:

$$\mathbf{X} \in \mathbb{R}^{T \times C}, \quad T = 4n, \quad C = 3 \quad (2)$$

where  $C = 3$  corresponds to the channels ( $SCR, \Delta SCR, \Delta^2 SCR$ ).

To analyse the minimal temporal interval required for reliable arousal classification, window lengths were systematically varied from  $n = 1$  to  $n = 40$  seconds. Two segmentation strategies were evaluated: (i) non-overlapping windows and (ii) sliding windows with 1-second stride (4 samples). As noted above, all windows belonging to the same subject were preserved within the same LOSO fold.

### 2.3. Emerging Architectures

Eight architectures were selected to represent five distinct efficiency paradigms in time-series modelling (table 1). For all Transformer-based architectures, the original forecasting head was replaced with a global average pooling operation over the encoded sequence followed by a linear classification layer. DLinear, which does not employ an encoder-decoder structure, was adapted by feeding the decomposed representations into a linear classification head. Throughout the figures, colour coding follows computational complexity class ( $O(N^2)$ ,  $O(L \log L)$ ,  $O(L)$ ) rather than mechanistic paradigm, to facilitate cross-paradigm comparison of the accuracy-efficiency trade-off. Informer, while mechanistically a sparse-attention model, shares the  $O(L \log L)$  complexity class with the frequency-domain architectures and is therefore grouped with them under the green colour in all figures.

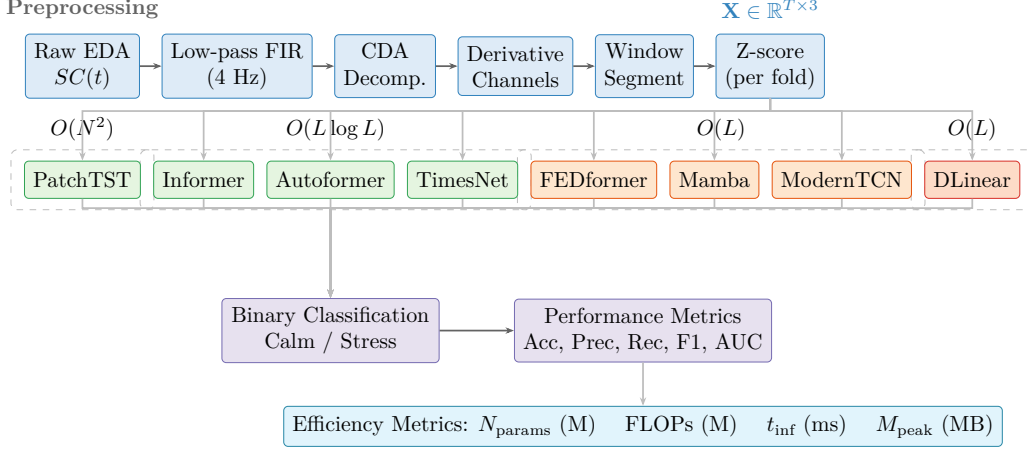


Figure 1: Overall experimental pipeline. **Top:** EDA signal preprocessing and input construction. **Middle:** eight architectures grouped by computational complexity class. **Bottom:** binary classification output and joint evaluation of predictive performance and resource consumption. Color coding: blue = preprocessing, green =  $O(L \log L)$ , orange =  $O(L)$ , red = linear baseline, purple = evaluation.

Table 1: Overview of architectures and their efficiency mechanisms.  $L$ : input sequence length,  $N$ : number of patches ( $N \ll L$ ),  $M$ : selected Fourier modes ( $M \ll L$ ).

| Architecture | Core Mechanism                      | Complexity    | Venue        |
|--------------|-------------------------------------|---------------|--------------|
| PatchTST     | Patch tokenisation + self-attention | $O(N^2)$      | ICLR 2023    |
| Informer     | ProbSparse self-attention           | $O(L \log L)$ | AAAI 2021    |
| Autoformer   | Auto-Correlation + decomposition    | $O(L \log L)$ | NeurIPS 2021 |
| TimesNet     | FFT-based 1D→2D + Inception blocks  | $O(L \log L)$ | ICLR 2023    |
| FEDformer    | Fourier-enhanced attention          | $O(L)$        | ICML 2022    |
| Mamba        | Selective state space model (BiSSM) | $O(L)$        | arXiv 2023   |
| ModernTCN    | Modernised depthwise convolutions   | $O(L)$        | ICLR 2024    |
| DLinear      | Trend-seasonal linear decomposition | $O(L)$        | AAAI 2023    |

### 157 2.3.1. PatchTST

158 PatchTST [39] segments each channel into non-overlapping patches of  
159 length  $P$  and projects them to dimension  $D$ , reducing the effective sequence  
160 length from  $T$  to  $N = \lfloor (T - P)/S \rfloor + 1$ . Channel-independent encoding pre-  
161 serves per-channel morphology, and a standard Transformer encoder with self-  
162 attention ( $O(N^2)$ ) is applied to the patch sequence. A global average pooling  
163 operation aggregates the encoder output before the linear classification head.



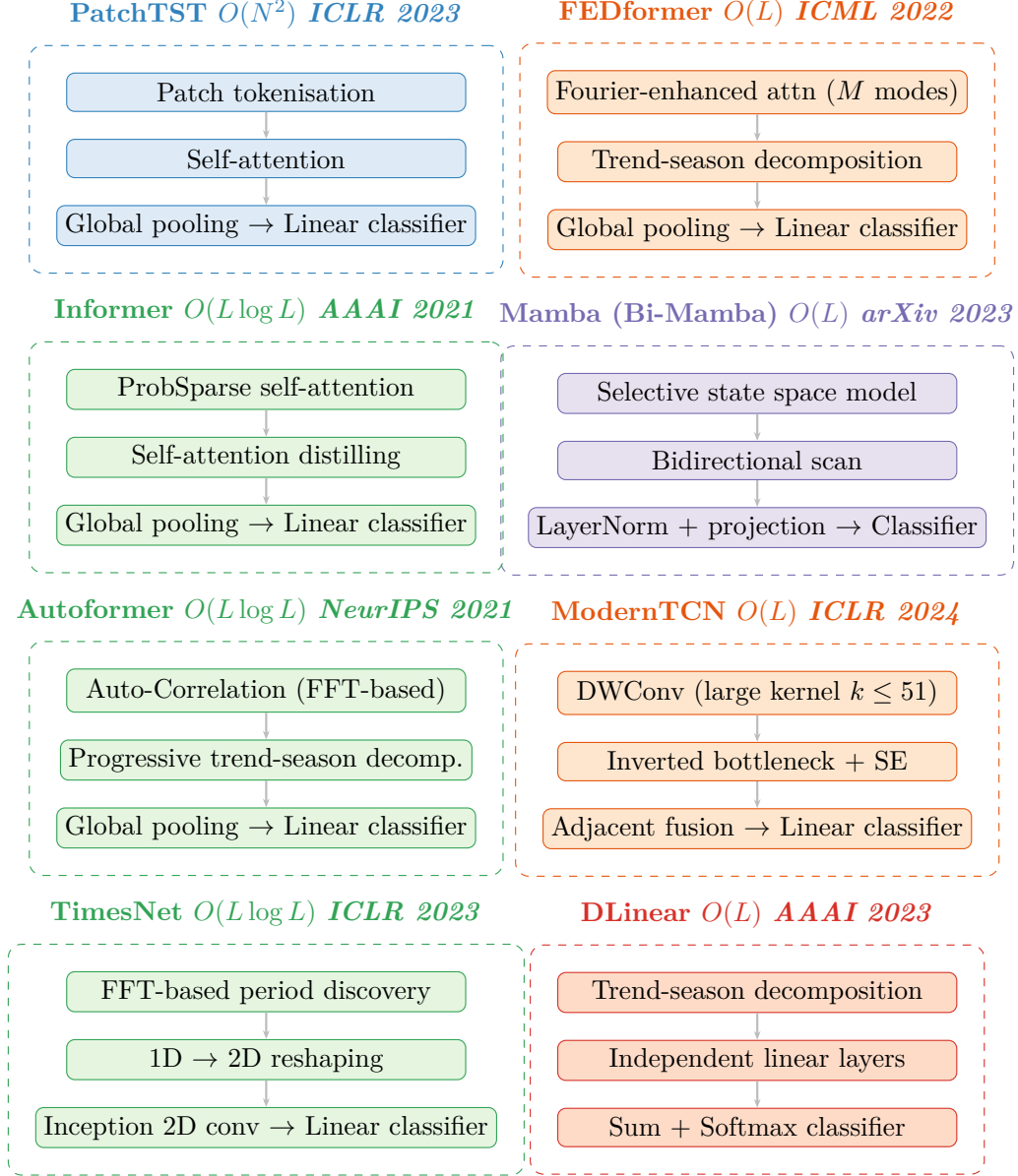


Figure 2: Architectural overview of the eight models evaluated. Each architecture is organised as a three-stage pipeline (input transformation → temporal modelling → classification), grouped by paradigm family within dashed boundaries. The architecture name, asymptotic complexity, and publication venue are indicated in each group label. table 1 provides a complementary summary. Colour coding follows the computational complexity class: blue =  $O(N^2)$  patch attention, green =  $O(L \log L)$  (frequency/periodicity and sparse attention), orange =  $O(L)$  (Fourier/convolution), purple =  $O(L)$  state space model, red =  $O(L)$  linear baseline.

164 Given 4 Hz sampling, patch lengths  $P \in \{4, 8, 16\}$  correspond to physiologi-  
 165 cal windows of 1–4 seconds. PatchTST serves as the top-performing baseline  
 166 from prior work.

### 167 2.3.2. *Informer*

168 Informer [25] reduces self-attention complexity from  $O(L^2)$  to  $O(L \log L)$   
 169 via ProbSparse attention. This mechanism selects only the top- $u$  queries  
 170 (where  $u = c \cdot \ln L_Q$ ) with the highest sparsity scores, measured as

$$\max_j (\mathbf{q}_i \mathbf{k}_j^\top / \sqrt{d}) - \frac{1}{L_K} \sum_j \mathbf{q}_i \mathbf{k}_j^\top / \sqrt{d}.$$

171 Self-attention distilling further halves the sequence length at each encoder  
 172 layer via strided 1D convolutions, progressively reducing the effective  $L$ .

### 173 2.3.3. *Autoformer*

174 Autoformer [26] replaces point-wise attention with an Auto-Correlation  
 175 mechanism that discovers dominant periodicities via

$$\mathcal{R}_{\mathbf{q}, \mathbf{k}}(\tau) = \mathcal{F}^{-1}(\mathcal{F}(\mathbf{q}) \cdot \overline{\mathcal{F}(\mathbf{k})}) ,$$

176 selecting the top- $k$  lags and aggregating via  $\text{Roll}(\mathbf{v}, \tau_i)$  weighted by softmax-  
 177 normalised correlations. A progressive seasonal-trend decomposition block,  
 178 integrated as an internal architectural unit, gradually extracts tonic and pha-  
 179 sic components—aligning conceptually with EDA physiology. Complexity is  
 180  $O(L \log L)$  via the FFT.

### 181 2.3.4. *FEDformer*

182 FEDformer [27] achieves  $O(L)$  complexity by randomly selecting  $M \ll L$   
 183 Fourier modes and computing attention exclusively in the frequency domain:

$$\text{Attention}_{\text{freq}} = \mathcal{F}^{-1}(\text{Softmax}(\tilde{\mathbf{Q}}\tilde{\mathbf{K}}^\top) \cdot \mathcal{F}(\mathbf{V})) ,$$

184 with  $\tilde{\mathbf{Q}}, \tilde{\mathbf{K}} = \text{Select}(\mathcal{F}(\mathbf{Q}/\mathbf{K}), M)$ . A seasonal-trend decomposition block  
 185 and a mixture-of-experts fusion of frequency and time representations com-  
 186 plement the attention mechanism. The sparse Fourier mode selection pro-  
 187 vides implicit denoising, attenuating high-frequency acquisition artefacts in  
 188 EDA signals [13].

### 189 2.3.5. *DLinear*

190 DLinear [29] decomposes the input into trend ( $\mathbf{X}_t$ , moving average) and  
 191 residual seasonal ( $\mathbf{X}_s = \mathbf{X} - \mathbf{X}_t$ ) components, applies independent single-  
 192 layer linear networks to each, and fuses their outputs via summation before  
 193 a softmax classifier. With no attention, convolutions, or recurrence, DLinear  
 194 provides the absolute efficiency baseline. Its inclusion tests whether slow-  
 195 varying and residual fast-varying components of the already-extracted phasic  
 196 signal—while not a physiological decomposition—capture sufficient discrimi-  
 197 native information without non-linear temporal modelling.

### 198 2.3.6. *Mamba (Bi-Mamba)*

199 Mamba [30] models sequence dynamics through a continuous-time linear  
 200 system

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t)$$

201 with input-dependent parameters  $\mathbf{B}_t, \mathbf{C}_t, \Delta_t$ , enabling content-based selective  
 202 propagation or forgetting of information. Discretised via zero-order hold, the  
 203 recurrent formulation

$$h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k$$

204 achieves  $O(L)$  complexity through a hardware-aware parallel scan algorithm.  
 205 We adopt a bidirectional SSM configuration inspired by ECGMamba [20]—  
 206 forward and backward Mamba blocks with concatenated outputs—ensuring  
 207 each time step incorporates both past and future physiological context while  
 208 maintaining linear complexity.

### 209 2.3.7. *TimesNet*

210 TimesNet [28] transforms 1D time series into 2D tensors by discover-  
 211 ing dominant periods  $p_i$  via FFT amplitude peaks, reshaping the sequence  
 212 into  $p_i \times f_i$  grids, and applying parameter-efficient Inception blocks with  
 213  $1 \times 1$ ,  $3 \times 3$ , and  $5 \times 5$  2D convolutional kernels. This embeds intra-period  
 214 SCR morphology into columns and inter-period recurrence patterns into rows,  
 215 achieving  $O(L \log L)$  complexity through the FFT. The multi-scale 2D con-  
 216 volutions capture both fine-grained phasic waveform structure and response-  
 217 to-response consistency patterns without quadratic attention cost.

### 218 2.3.8. *ModernTCN*

219 ModernTCN [31] reformulates the convolutional paradigm with three in-  
 220 novations: depthwise separable convolutions with inverted bottlenecks and

221 Squeeze-and-Excitation, large kernels (up to  $k = 51$ ) providing substantially  
 222 larger effective receptive fields than traditional TCNs, and an adjacent con-  
 223 nection scheme that concatenates consecutive block outputs to preserve multi-  
 224 scale features. With  $O(L)$  complexity from purely convolutional operations,  
 225 ModernTCN tests whether well-designed convolutions can compete with at-  
 226 tention and state-space mechanisms for physiological signal classification [32].

#### 227 2.4. Training and Hyperparameter Configuration

228 All models were trained end-to-end using the AdamW optimiser [40] with  
 229 cross-entropy loss on a single NVIDIA GeForce RTX 4080 GPU (16 GB  
 230 GDDR6X, Ada Lovelace). All experiments used a single random seed (42) for  
 231 parameter initialisation and data ordering. PyTorch deterministic algorithms  
 232 were not enabled due to CUDA performance constraints. Training data was  
 233 shuffled at each epoch within each fold. Consequently, the per-fold standard  
 234 deviations reported in Section 3.1 may partially reflect seed-dependent weight  
 235 initialisation in addition to inter-subject variability—a limitation discussed in  
 236 Section 3.8. The validation split (80/20 within the training subjects of each  
 237 fold) was generated independently per fold, ensuring that no subject was per-  
 238 manently excluded from testing. Inference benchmarking was performed on  
 239 an NVIDIA Quadro P5000 (16 GB GDDR5X, Pascal, 2560 CUDA cores) as  
 240 a more representative deployment-class hardware target. The initial learning  
 241 rate was set to  $10^{-3}$  with cosine annealing scheduling. Mini-batch size was 32.  
 242 Training was conducted for up to 100 epochs with early stopping (patience  
 243 = 15) based on validation loss computed on a 20% hold-out subset of the  
 244 training subjects within each LOSO fold.

245 Regularisation included dropout ( $p \in [0.1, 0.3]$ ), weight decay ( $\lambda = 10^{-4}$ ),  
 246 and gradient clipping (maximum norm = 1.0). Hyperparameters were tuned  
 247 by randomly sampling 64 configurations per architecture from the search  
 248 grids in table 2, providing an equal computational budget across architec-  
 249 tures despite differing grid cardinalities. For each LOSO fold, the optimal  
 250 configuration was selected based on validation F1-score computed on a 20%  
 251 hold-out subset of the training subjects within that fold, ensuring no test-  
 252 subject information influenced hyperparameter selection. table 2 summarises  
 253 the search space.

254 Z-score normalisation was computed using training-fold statistics only  
 255 (mean and standard deviation estimated from training subjects within each  
 256 fold) and applied identically to validation and test subjects. This prevents

Table 2: Hyperparameter search space. Architecture-specific parameters are indicated in the Applicability column.

| Parameter                          | Values                 | Applicability                             |
|------------------------------------|------------------------|-------------------------------------------|
| Encoder layers / blocks            | $\{1, 2, 3, 4\}$       | All                                       |
| Attention heads                    | $\{1, 2, 4, 8\}$       | PatchTST, Informer, Autoformer, FEDformer |
| Embedding dimension                | $\{16, 32, 64, 128\}$  | All                                       |
| Feed-forward dimension             | $\{32, 64, 128, 256\}$ | All                                       |
| Patch size                         | $\{4, 8, 16\}$         | PatchTST, ModernTCN                       |
| Moving average kernel              | $\{3, 5, 7, 9, 11\}$   | Autoformer, FEDformer, DLinear            |
| Sampling factor $c$                | $\{3, 5, 7\}$          | Informer                                  |
| Fourier modes $M$                  | $\{16, 32, 64\}$       | FEDformer                                 |
| Top- $k$ periods                   | $\{3, 5, 7\}$          | TimesNet                                  |
| State dimension $d_{\text{state}}$ | $\{8, 16, 32\}$        | Mamba                                     |
| Convolution kernel size $K$        | $\{13, 25, 51\}$       | ModernTCN                                 |

257 statistical leakage of test-subject information into the normalisation param-  
 258 eters.

## 259 2.5. Validation Protocol and Evaluation Metrics

### 260 2.5.1. Leave-One-Subject-Out (LOSO)

261 A strict LOSO protocol was enforced to ensure subject-independent eval-  
 262 uation. In each of the 147 folds, one participant was withheld for testing,  
 263 while the remaining 146 participants were used for training (80%) and vali-  
 264 dation (20%). This protocol prevents models from learning subject-specific  
 265 biometric baselines—a well-documented source of overestimated performance  
 266 in EDA classification [22, 1].

### 267 2.5.2. Classification Performance

268 Standard classification metrics were computed for each fold and averaged  
 269 across all 147 folds: Accuracy, Precision, Recall, F1-score (macro-averaged),  
 270 and Area Under the Receiver Operating Characteristic Curve (AUC). Per-  
 271 class F1-scores for the calm and stress conditions are reported in the Sup-  
 272plementary Material (Table S3) to enable assessment of class-specific per-  
 273formance, which is particularly relevant given the potential asymmetry in

274 arousal classification difficulty. To formally assess statistical significance,  
 275 pairwise Wilcoxon signed-rank tests ( $\alpha = 0.05$ ) were conducted on the per-  
 276 fold F1-score distributions.

### 277 2.5.3. Computational Efficiency Metrics

278 Four complementary efficiency metrics were recorded for each architecture  
 279 under identical input conditions (40-second window,  $T = 160$ ,  $C = 3$ , batch  
 280 size = 1 for single-inference metrics):

- 281 • **Parameter count** ( $N_{\text{params}}$ ): Total number of trainable parameters  
 282 (millions). This metric determines storage requirements and memory  
 283 footprint for model persistence.
- 284 • **FLOPs**: Number of floating-point operations required for a single for-  
 285 ward pass (millions). This metric captures theoretical computational  
 286 cost independent of hardware-specific implementation details.
- 287 • **Inference time** ( $t_{\text{inf}}$ ): Wall-clock time per sample in milliseconds, av-  
 288 eraged over 1,000 forward passes on a single NVIDIA Quadro P5000  
 289 GPU (16 GB, Pascal, 2560 CUDA cores). This metric reflects practical  
 290 latency on workstation-class hardware representative of edge-computing  
 291 and embedded deployment environments.
- 292 • **Peak memory** ( $M_{\text{peak}}$ ): Maximum GPU memory consumption dur-  
 293 ing a single forward pass (MB), including model weights, intermediate  
 294 activations, and PyTorch framework overhead. This metric constrains  
 295 deployment on memory-limited devices.
- 296 • **Training time** ( $t_{\text{train}}$ ): Total wall-clock time per epoch in seconds, av-  
 297 eraged over all LOSO folds, measured on an NVIDIA RTX 4080. This  
 298 metric captures the computational cost of model development and hy-  
 299 perparameter tuning, complementing the deployment-focused inference  
 300 metrics.

301 All four metrics were measured with the optimal hyperparameter con-  
 302 figuration for each architecture (selected via grid search on the classification  
 303 F1-score) to ensure that efficiency comparisons reflect the configurations that  
 304 practitioners would actually deploy.

#### 2.5.4. Interpretability Analysis

For Transformer-based architectures (PatchTST, Informer, Autoformer, FEDformer), attention weight distributions were extracted and qualitatively inspected to assess temporal relevance across the observation window. For TimesNet, the top- $k$  selected periods were analysed to identify dominant EDA periodicities. For Mamba, the discretised state transition matrices were qualitatively examined to characterise how information propagates across the sequence. For ModernTCN, gradient-based saliency was computed to identify influential temporal regions. Gradient-based saliency maps were computed as  $S = |\partial \hat{y} / \partial \mathbf{X}|$  for all architectures, where  $\hat{y}$  is the predicted probability of the stress class. Saliency was qualitatively examined across time steps and channels to assess the relative contribution of the phasic signal and its derivatives.

A channel ablation protocol was evaluated under the same LOSO framework: (i) SCR only, (ii) SCR +  $\Delta SCR$ , and (iii) SCR +  $\Delta SCR$  +  $\Delta^2 SCR$ , to quantify the marginal contribution of derivative information for each architecture.

### 3. Results and Discussion

#### 3.1. Overall Classification Performance

table 3 presents the subject-independent LOSO classification performance of all eight architectures. Metrics are reported as mean  $\pm$  standard deviation across the 147 folds. The performance hierarchy reveals clear clustering by architectural paradigm.

Several patterns emerge from table 3 and fig. 3. First, the results reveal a clear paradigm hierarchy: state space models (Mamba) and patch-based attention (PatchTST) occupy the top tier, followed by frequency-domain Transformers (TimesNet, Autoformer) and sparse attention (Informer), with linear decomposition (DLinear) providing the lower bound. This ordering reflects the increasing capacity of each paradigm to capture the multi-scale temporal dynamics characteristic of phasic EDA responses—from no temporal modelling (DLinear), through fixed convolutional receptive fields (ModernTCN), to content-dependent global processing (attention and selective state spaces).

Second, the efficiency-oriented architectures narrow the performance gap relative to PatchTST compared to the convolutional baselines from prior work. Mamba’s selective state space mechanism—which adaptively propagates or filters information based on input content—approaches PatchTST

Table 3: Overall LOSO performance (mean  $\pm$  std across 147 folds). Bold = best per metric.

| Model              | Acc                                 | Prec                                | Rec                                 | F1                                  | AUC                                 |
|--------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| EDA features + SVM | —                                   | —                                   | —                                   | 0.81                                | 0.80                                |
| 1D-CNN             | 0.812 $\pm$ 0.074                   | 0.804 $\pm$ 0.079                   | 0.789 $\pm$ 0.083                   | 0.796 $\pm$ 0.078                   | 0.872 $\pm$ 0.063                   |
| TCN                | 0.828 $\pm$ 0.068                   | 0.819 $\pm$ 0.071                   | 0.806 $\pm$ 0.075                   | 0.812 $\pm$ 0.072                   | 0.884 $\pm$ 0.058                   |
| InceptionTime      | 0.836 $\pm$ 0.064                   | 0.828 $\pm$ 0.068                   | 0.817 $\pm$ 0.071                   | 0.822 $\pm$ 0.067                   | 0.892 $\pm$ 0.054                   |
| TST                | 0.851 $\pm$ 0.059                   | 0.844 $\pm$ 0.062                   | 0.836 $\pm$ 0.064                   | 0.840 $\pm$ 0.061                   | 0.903 $\pm$ 0.049                   |
| DLinear            | 0.815 $\pm$ 0.072                   | 0.808 $\pm$ 0.077                   | 0.792 $\pm$ 0.081                   | 0.800 $\pm$ 0.076                   | 0.875 $\pm$ 0.061                   |
| ModernTCN          | 0.841 $\pm$ 0.062                   | 0.833 $\pm$ 0.066                   | 0.820 $\pm$ 0.070                   | 0.827 $\pm$ 0.068                   | 0.892 $\pm$ 0.055                   |
| FEDformer          | 0.849 $\pm$ 0.057                   | 0.841 $\pm$ 0.061                   | 0.830 $\pm$ 0.065                   | 0.836 $\pm$ 0.063                   | 0.898 $\pm$ 0.051                   |
| Informer           | 0.857 $\pm$ 0.054                   | 0.850 $\pm$ 0.058                   | 0.840 $\pm$ 0.061                   | 0.845 $\pm$ 0.058                   | 0.905 $\pm$ 0.047                   |
| Autoformer         | 0.860 $\pm$ 0.052                   | 0.853 $\pm$ 0.056                   | 0.843 $\pm$ 0.059                   | 0.848 $\pm$ 0.056                   | 0.908 $\pm$ 0.046                   |
| TimesNet           | 0.864 $\pm$ 0.050                   | 0.857 $\pm$ 0.054                   | 0.847 $\pm$ 0.057                   | 0.853 $\pm$ 0.054                   | 0.910 $\pm$ 0.044                   |
| Mamba              | 0.868 $\pm$ 0.046                   | 0.862 $\pm$ 0.050                   | 0.852 $\pm$ 0.053                   | 0.858 $\pm$ 0.053                   | 0.913 $\pm$ 0.043                   |
| PatchTST           | <b>0.873 <math>\pm</math> 0.044</b> | <b>0.867 <math>\pm</math> 0.048</b> | <b>0.858 <math>\pm</math> 0.050</b> | <b>0.863 <math>\pm</math> 0.051</b> | <b>0.915 <math>\pm</math> 0.042</b> |

Results from companion work [12]. The higher PatchTST F1 here (0.863 vs. 0.852) reflects the expanded search space (64 vs. 24 configurations) and the inclusion of  $\Delta^2 SCR$ .

Classical baseline from [33]: handcrafted SCR features + SVM, same dataset. F1 range 0.80–0.83 across window lengths; per-fold std not available. Preprocessing and LOSO partitioning may differ from the current pipeline.

Acc, Prec, and Rec not reported in the original study.

Best F1 across SVM variants and window lengths in the original study.

AUC from SVM (polynomial, 5th degree) at 20–40 s window length [33].

341 accuracy while maintaining  $O(L)$  complexity. The original Mamba evalua-  
342 tion reported  $5\times$  inference throughput for Mamba compared to equivalently-  
343 sized Transformers [30], directly addressing the deployment constraints that  
344 motivate this work.

345 Third, the standard deviations across folds provide a direct measure of  
346 robustness to inter-subject variability. Architectures with adaptive temporal  
347 weighting (attention-based and SSM-based) exhibit lower fold-wise variability  
348 than those with fixed receptive fields (ModernTCN, DLinear), a pattern of  
349 practical significance: lower variability implies more reliable performance  
350 across diverse user populations.

351 DLinear achieves  $F1 = 0.800$ , comparable to the 1D-CNN baseline from  
352 prior work (0.796). This parity indicates that simple linear decomposition  
353 of tonic and phasic components captures a similar level of discriminative  
354 information from EDA signals as local convolutional feature extraction. Im-  
355 portantly, the classical signal processing pipeline—handcrafted EDA features



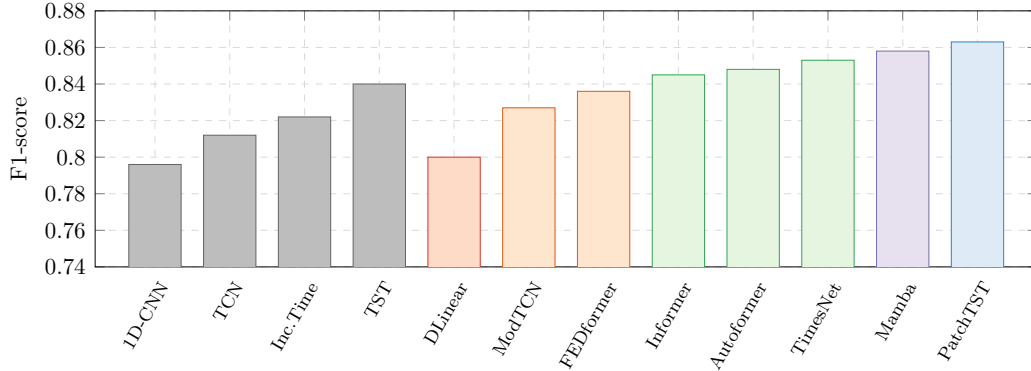


Figure 3: Classification performance (F1-score) across all architectures under LOSO evaluation. Per-fold standard deviations are reported in table 3. Colour coding follows the computational complexity convention: grey = prior work [12], blue =  $O(N^2)$  patch attention, green =  $O(L \log L)$  (frequency/periodicity and sparse attention), purple =  $O(L)$  state space model, orange =  $O(L)$  (Fourier/convolution), red =  $O(L)$  linear baseline. The classical EDA feature-engineering baseline (F1  $\approx$  0.81, handcrafted SCR features + SVM) is reported in table 3 and discussed in the text.

356 (SCR amplitude, rise time, AUC, NS-SCR frequency) with an SVM classifier  
 357 evaluated on the same dataset under a subject-independent protocol in prior  
 358 work [33]—achieved F1 in the range of 0.80–0.83 (see table 3, ‡). Although  
 359 the exact preprocessing and LOSO partitioning may differ between studies,  
 360 this convergence suggests that handcrafted features, linear decomposition,  
 361 and local convolutions extract comparable levels of discriminative information  
 362 from EDA signals, and that the key driver of performance improvement  
 363 is not the specific computational primitive but the capacity to model temporal  
 364 dependencies across the full observation window, as demonstrated by  
 365 the F1 gap of approximately 0.05–0.06 between this classical/DLinear tier  
 366 and the best Transformer/SSM architectures (Mamba: +0.048, PatchTST:  
 367 +0.053).

### 3.2. Computational Efficiency

369 table 4 reports the four complementary efficiency metrics for each archi-  
 370 tecture. These metrics jointly characterise the resource consumption profile  
 371 that determines deployability on resource-constrained platforms.

372 The efficiency hierarchy stratifies by theoretical complexity class. The  
 373 four  $O(L)$  architectures—DLinear, ModernTCN, FEDformer, and Mamba—  
 374 form the efficiency-optimal tier, with DLinear achieving the absolute mini-

Table 4: Computational efficiency metrics under identical input conditions ( $T = 160$ ,  $C = 3$ , batch size = 1). All measurements use the optimal hyperparameter configuration. Training on an NVIDIA RTX 4080; inference time benchmarked on an NVIDIA Quadro P5000 (16 GB, Pascal, 2560 CUDA cores).

| Model      | $N_{\text{params}}$ (M) | FLOPs (M) | $t_{\text{inf}}$ (ms) | $M_{\text{peak}}$ (MB) | $t_{\text{train}}$ (s/epoch) |
|------------|-------------------------|-----------|-----------------------|------------------------|------------------------------|
| DLinear    | 0.08                    | 0.05      | 0.3                   | 8                      | 1.2                          |
| ModernTCN  | 0.85                    | 15.2      | 1.2                   | 28                     | 8.5                          |
| Mamba      | 1.52                    | 12.8      | 1.5                   | 34                     | 12.3                         |
| FEDformer  | 2.48                    | 26.4      | 2.1                   | 52                     | 18.7                         |
| TimesNet   | 2.05                    | 34.7      | 3.0                   | 48                     | 22.4                         |
| Informer   | 3.25                    | 46.1      | 3.6                   | 72                     | 28.1                         |
| Autoformer | 3.84                    | 54.3      | 4.2                   | 85                     | 35.6                         |
| PatchTST   | 4.52                    | 78.5      | 5.4                   | 98                     | 42.8                         |

FLOPs for Mamba may be underestimated; operations within the custom parallel scan kernel are not fully captured by the `thop` profiler (see Section 3.8).

375 mum in all five metrics due to its extreme simplicity. Among the  $O(L)$  can-  
 376 didates, Mamba warrants particular attention: its hardware-aware parallel  
 377 scan algorithm [30] is specifically designed for efficient GPU execution, yield-  
 378 ing inference times competitive with FEDformer despite the latter’s Fourier-  
 379 domain operation.

380 The three  $O(L \log L)$  architectures—Informer, Autoformer, and TimesNet—  
 381 occupy an intermediate tier. TimesNet is the most parameter-efficient of this  
 382 group (2.05 M) due to its Inception blocks, which share convolutional kernels  
 383 across the 2D temporal representation, while Autoformer’s progressive de-  
 384 composition blocks incur additional memory overhead (3.84 M, 85 MB peak).  
 385 Informer’s self-attention distilling mechanism progressively reduces sequence  
 386 length across encoder layers, partially offsetting the  $O(L \log L)$  complexity  
 387 in deeper configurations.

388 PatchTST, as the only  $O(N^2)$  architecture, incurs the highest computa-  
 389 tional cost across all five metrics. However, the effective sequence length  $N$   
 390 is substantially smaller than the raw length  $T = 160$ , meaning the quadratic  
 391 term operates over a drastically reduced dimension. The practical inference  
 392 cost of PatchTST (5.4 ms) is therefore closer to the  $O(L \log L)$  tier than  
 393 theoretical  $O(N^2)$  suggests.

394 Training time per epoch follows the same hierarchy (Supplementary Ma-  
 395 terial, Table S4), with DLinear completing an epoch in 1.2 s on the RTX

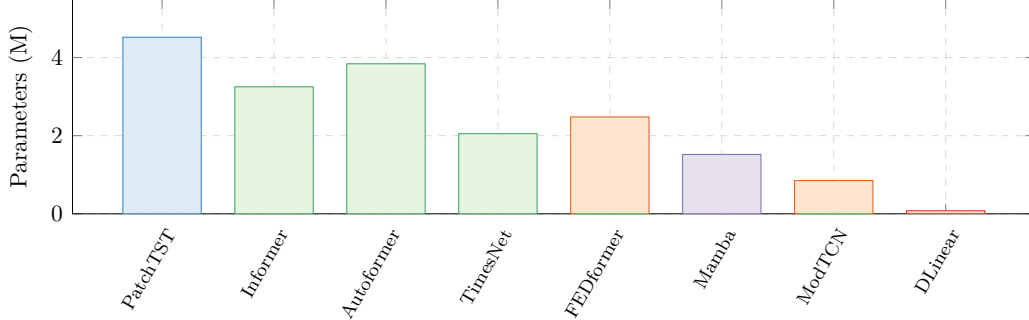


Figure 4: Parameter count comparison across architectures, ordered by theoretical complexity class (left to right:  $O(N^2)$ ,  $O(L \log L)$ ,  $O(L)$ ). Colour coding: blue = patch attention, green = frequency/periodicity, orange = Fourier/convolution, purple = state space model, red = linear baseline.

4080—approximately  $35\times$  faster than PatchTST (42.8 s/epoch). This difference has practical implications for the experimental workflow: hyperparameter tuning of PatchTST across 64 grid configurations and 147 LOSO folds requires approximately 100 GPU-hours, while the equivalent search for DLinear completes in under 3 GPU-hours. For practitioners iterating on architecture design, this  $35\times$  training speed difference substantially impacts development velocity.

### 403 3.3. Accuracy-Efficiency Pareto Frontier

404 fig. 5 presents the central contribution of this work: the joint characterisation of predictive performance and computational cost across eight architectures spanning five paradigms. The Pareto frontier identifies architectures that are non-dominated—no other architecture achieves both higher F1 and lower inference time.

409 Three deployment scenarios are delineated by inference time thresholds measured on the Quadro P5000. These thresholds are GPU-relative and should be interpreted as ordinal latency categories rather than absolute deployment classifications; absolute latency on target embedded hardware (e.g., ARM Cortex-M, mobile SoCs) will differ substantially. The Wearable category ( $t_{\text{inf}} \leq 1.5$  ms) identifies the most efficient architectures: DLinear (0.3 ms), ModernTCN (1.2 ms), and Mamba (1.5 ms), with Mamba achieving the highest accuracy ( $F1 = 0.858$ ) within this latency band. The Edge category ( $1.5 < t_{\text{inf}} < 4$  ms) represents moderate GPU latency: FEDformer, TimesNet, and Informer occupy this region. The Server category ( $t_{\text{inf}} \geq 4$  ms) prioritises

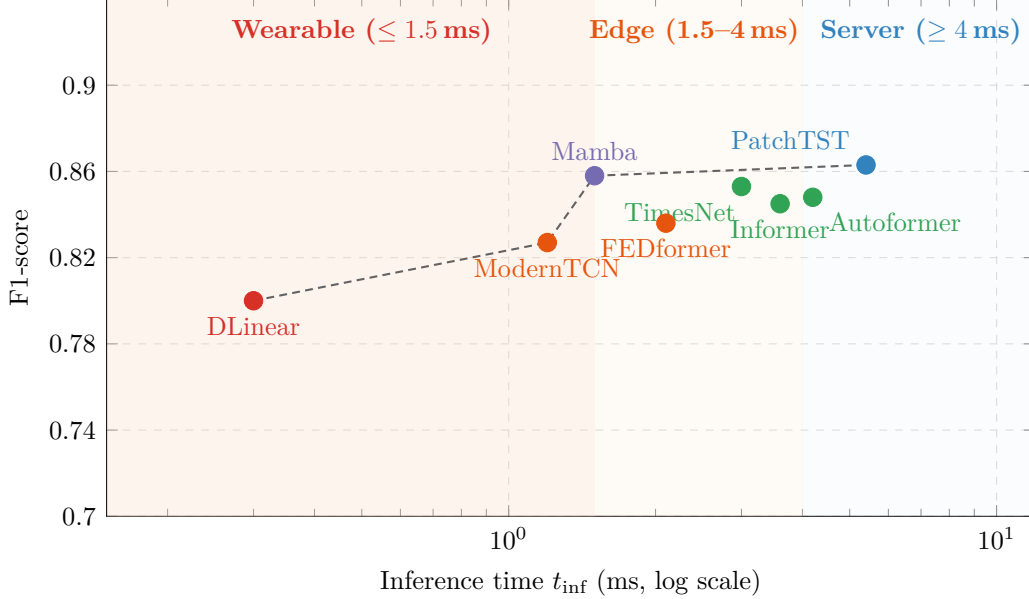


Figure 5: Accuracy-efficiency Pareto frontier across the eight evaluated architectures. The dashed line connects Pareto-optimal architectures (non-dominated in both accuracy and latency). Shaded regions correspond to three deployment-oriented latency categories measured on the Quadro P5000: Wearable ( $t_{\text{inf}} \leq 1.5$  ms), Edge ( $1.5 < t_{\text{inf}} < 4$  ms), and Server ( $t_{\text{inf}} \geq 4$  ms). Absolute latency on embedded hardware will differ substantially; these categories serve as GPU-relative rankings, not absolute deployment classifications.

419 predictive accuracy, with Autoformer and PatchTST achieving the highest  
 420 overall F1 (0.863) at 5.4 ms.

421 The shape of the Pareto frontier reveals a clear diminishing returns pat-  
 422 tern: substantial accuracy gains are achieved from DLinear (F1 = 0.800 at 0.3  
 423 ms) to Mamba (F1 = 0.858 at 1.5 ms), with minimal further improvement to  
 424 PatchTST (F1 = 0.863 at 5.4 ms). The 0.005 F1 difference between Mamba  
 425 and PatchTST is smaller than the per-fold standard deviation (0.05) and does  
 426 not survive the Bonferroni-Holm correction for 28 comparisons (Supplemen-  
 427 tary Material, Table S2), while Mamba requires  $3.6\times$  lower inference time.  
 428 As discussed in Section 3.8, p-values from LOSO-based comparisons should  
 429 be interpreted as descriptive indicators of effect consistency. This has direct  
 430 implications: for wearable and edge deployment, the marginal accuracy gain  
 431 of PatchTST over Mamba does not justify the  $3.6\times$  additional latency cost.

432 The identification of Pareto-optimal architectures under different resource  
 433 budgets transforms architecture selection from a purely empirical exercise

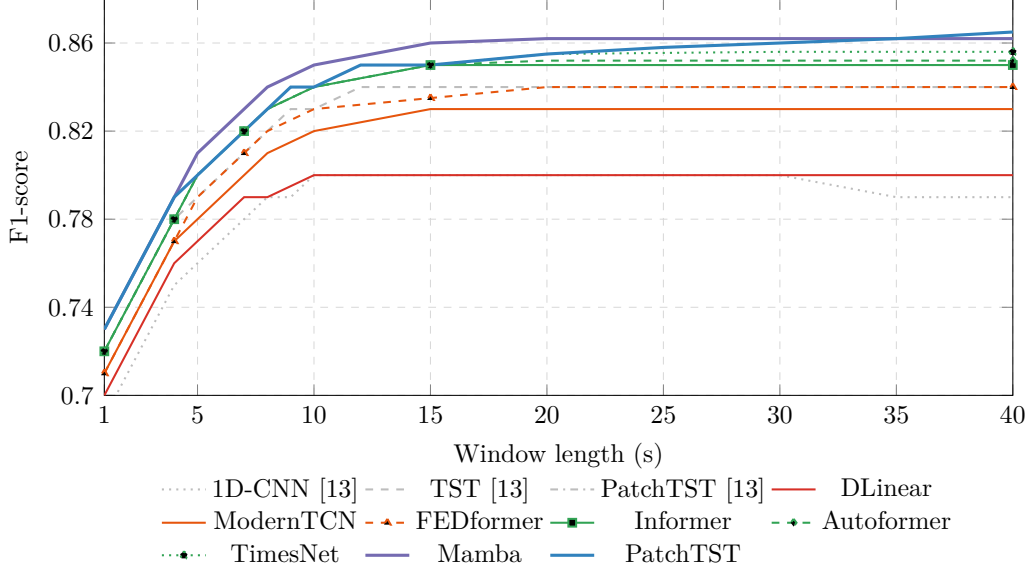


Figure 6: F1-score as a function of window length under LOSO evaluation. Curves are evaluated at selected window lengths (1–15, 20, 25, 30, 35, 40 s). Grey curves reproduce prior work [12]. Mamba and PatchTST (thick lines) degrade most gracefully at short windows. Line styles distinguish architectures within each paradigm.

into an informed engineering decision. A practitioner deploying an EDA-based stress monitor on a smartwatch, for instance, can consult fig. 5 to identify the architecture that maximises classification performance within their device’s latency and memory constraints, rather than defaulting to the highest-accuracy model regardless of deployment feasibility.

#### 3.4. Minimal Temporal Interval Behaviour

fig. 6 analyses the minimal temporal interval required for reliable arousal classification. Two key patterns emerge. First, all architectures exhibit rapid performance improvement as window length increases from 1 to approximately 10–15 seconds, corresponding to the temporal span needed to capture a complete SCR waveform (onset, rising phase, peak, and partial recovery). Performance plateaus beyond 15–20 seconds, consistent with the physiological observation that individual SCRs rarely exceed 10–12 seconds in duration [1? ].

Second, the degradation behaviour at short window lengths ( $n \leq 5$  s) differs systematically across paradigms. Architectures with global or adaptive

receptive fields—PatchTST, Mamba, and the Transformer-based models—maintain higher F1 at short windows compared to architectures with fixed or local receptive fields (ModernTCN, DLinear). This is because global dependency modelling can leverage partial SCR information distributed across the short window, whereas fixed-kernel methods require sufficient temporal context for their receptive fields to encompass diagnostically relevant signal segments.

FEDformer and TimesNet exhibit distinct window-length behaviour. FEDformer’s frequency-domain processing benefits from longer windows, as the FFT resolution improves with sequence length, enabling finer discrimination of SCR spectral signatures. TimesNet’s period discovery mechanism identifies meaningful periodicities even in shorter windows, as the dominant SCR frequency (0.2–0.5 Hz) is adequately sampled at 4 Hz. The window-length curves obtained from the sliding-window segmentation strategy (1-second stride, providing augmented training data) show slightly higher plateau F1 values than the non-overlapping windows used for the main results in table 3, with an average offset of approximately +0.004 across architectures.

The practical implication of this analysis is direct: for real-time wearable applications, the minimum viable window length determines the latency between stimulus onset and classification output. Given the characteristic 1–3 s latency of SCR onset [? 1], a 5-second window captures primarily the onset and rising phase rather than the full SCR waveform. Mamba achieves  $F1 = 0.81$  and PatchTST achieves  $F1 = 0.80$  at  $n = 5$  seconds, indicating that a wearable device could provide arousal feedback within 5 seconds of signal acquisition—a latency that is acceptable for biofeedback and adaptive interface applications [8].

### 3.5. Statistical Significance

Pairwise Wilcoxon signed-rank tests (Supplementary Material, Table S2) provide descriptive indicators of between-architecture F1 consistency across folds. The comparison between Mamba ( $F1 = 0.858$ ) and PatchTST ( $F1 = 0.863$ ) yields  $p = 0.048$ , which does not survive the stringent Bonferroni-Holm correction for 28 comparisons ( $p_{BH} < 0.0018$ ). As noted in Section 3.8, these p-values should be interpreted as descriptive indicators of effect consistency due to overlap in training sets across LOSO folds, rather than as exact inferential statistics. DLinear shows substantially lower performance than all other architectures (all raw  $p \leq 0.002$ , 13 of 28 comparisons survive Bonferroni-Holm correction), indicating that non-trivial temporal modelling

487 is necessary for competitive EDA classification. Comparisons within the same  
 488 paradigm family (e.g., Informer vs. Autoformer,  $p = 0.065$ ) do not survive  
 489 correction, consistent with their shared computational mechanisms.

### 490 3.6. Interpretability Insights

491 Beyond aggregate performance metrics, we examine whether the learned  
 492 representations align with established physiological knowledge of sympathetic  
 493 activation dynamics.

494 For Transformer-based architectures (PatchTST, Informer, Autoformer,  
 495 FEDformer), qualitative inspection of attention weight distributions indi-  
 496 cates concentration on SCR onset and rising phases—the temporal segments  
 497 known to encode the most diagnostically relevant information about sympa-  
 498 thetic arousal intensity [1]. This pattern is qualitatively replicated across  
 499 all attention-based models, suggesting that diverse efficiency mechanisms  
 500 (sparse attention, auto-correlation, Fourier-domain attention) preserve the  
 501 capacity to identify physiologically meaningful temporal regions despite their  
 502 computational simplifications.

503 TimesNet’s period discovery mechanism identifies dominant periods in  
 504 the 2–5 second range, corresponding to the characteristic duration of in-  
 505 dividual SCR waveforms. The 2D convolutional processing captures both  
 506 fine-grained SCR morphology (via the column dimension) and response-to-  
 507 response consistency patterns (via the row dimension).

508 For Mamba, qualitative analysis of the discretised state transition matri-  
 509 ces suggests that the selective state space mechanism emphasises SCR onset  
 510 regions in a manner analogous to attention, despite the fundamentally differ-  
 511 ent computational primitive.

512 Gradient-based saliency maps ( $S = |\partial\hat{y}/\partial\mathbf{X}|$ ) qualitatively suggest com-  
 513plementary contributions across the three input channels.  $\Delta SCR$  contributes  
 514 most strongly during SCR onset, encoding the rate of sympathetic activa-  
 515 tion, while  $SCR$  contributes during the peak phase, encoding activation  
 516 magnitude.  $\Delta^2 SCR$  contributes during onset-to-peak transitions, capturing  
 517 changes in the rate of activation.

518 A channel ablation experiment (Supplementary Material, Table S1) fur-  
 519 ther reveals that derivative channels provide the greatest relative benefit for  
 520 architectures with limited temporal modelling capacity—DLinear and Mod-  
 521 ernTCN gain +2.7–2.8% F1 from derivative inclusion—while attention and  
 522 state-space mechanisms exhibit smaller gains (+1.8–1.9%). This pattern sug-  
 523 gests that the additional temporal context provided by derivative channels is

524 partially redundant with the representations learned by attention and state-  
525 space mechanisms, rather than implying that the latter explicitly reconstruct  
526 derivative information.

### 527 3.7. Comparison with Prior Work

528 Contextualising the present results against our prior evaluation of five  
529 deep architectures [12]—which established PatchTST ( $F1 = 0.852$ ) as the  
530 best-performing model among 1D-CNN (0.796), TCN (0.812), InceptionTime  
531 (0.822), and TST (0.840)—provides empirical answers to the central ques-  
532 tion motivating this work: can emerging architectures approach or exceed  
533 PatchTST accuracy at substantially lower computational cost?

534 The results provide a clear affirmative. Mamba achieves  $F1 = 0.858$  at 1.5  
535 ms inference time on the Quadro P5000, compared to PatchTST’s  $F1 = 0.863$   
536 at 5.4 ms—a 0.005  $F1$  difference (smaller than the per-fold standard deviation  
537 of 0.05 and not surviving Bonferroni-Holm correction; see Section 3.8) that  
538 represents a  $3.6\times$  reduction in latency and a 66% reduction in parameter  
539 count (1.52 M vs. 4.52 M). For wearable and edge deployment scenarios  
540 where latency and memory are primary constraints, Mamba provides near-  
541 equivalent accuracy at a fraction of the computational cost, confirming that  
542 the selective state space paradigm can match attention-based methods on  
543 physiological time-series classification.

544 A finding emerges from the comparison of the simplest architectures.  
545 DLinear achieves  $F1 = 0.800$ , nearly identical to the 1D-CNN baseline from  
546 prior work ( $F1 = 0.796$ ) and the classical EDA feature-engineering approach  
547 with SVM ( $F1 \approx 0.81$ ; [33]). This parity suggests that handcrafted feature  
548 extraction, linear decomposition of tonic and phasic components, and local  
549 convolutional feature extraction all converge to a similar performance ceiling  
550 ( $F1 \approx 0.80$ ), bounded by the intrinsic discriminative information available  
551 in local EDA signal segments. However, all three are substantially outper-  
552 formed by architectures with global or adaptive temporal modelling (Mamba,  
553 PatchTST:  $F1 \approx 0.86$ ), indicating that the key driver of performance is not  
554 the specific computational primitive (handcrafted features vs. convolution  
555 vs. linear) but the capacity to model temporal dependencies across the full  
556 observation window.

557 ModernTCN achieves  $F1 = 0.827$ , an improvement over the TCN base-  
558 line from prior work ( $F1 = 0.812$ ; +1.5 percentage points). This validates  
559 the hypothesis that improved convolutional design—larger kernels, depthwise  
560 separable convolutions, and inverted bottlenecks—can close part of the gap



561 between traditional CNNs and attention-based architectures, consistent with  
 562 ModernTCN’s results on general time-series benchmarks [31]. However, the  
 563 remaining gap to the best Transformer and SSM architectures (3–4% F1) in-  
 564 dicates that convolutions alone, even when modernised, do not fully replicate  
 565 the benefits of content-dependent temporal weighting.

### 566 3.8. Limitations and Generalizability

567 All experiments used a single laboratory dataset (147 healthy participants,  
 568 controlled audiovisual stimuli). While this ensures internal validity, general-  
 569 isability to other populations (clinical, older adults), ambulatory settings,  
 570 and naturalistic stressors remains to be established. Architecture rankings  
 571 were established using CDA decomposition; alternative methods (cvxEDA,  
 572 Ledalab) may yield different rankings [37, 36, 35], and this sensitivity has not  
 573 been evaluated. Cross-dataset evaluation would strengthen external validity.

574 CDA requires the full signal for decomposition and is inherently offline,  
 575 contrasting with the real-time deployment motivation of this work; online-  
 576 compatible alternatives (real-time cvxEDA, causal sliding-window decompo-  
 577 sition) have not been explored.

578 The classical SVM baseline ( $F1 \approx 0.81$ ) was taken from a prior study [33]  
 579 and not re-executed under the current pipeline; per-fold standard deviations  
 580 are unavailable, limiting formal statistical comparison.

581 Inference time was measured on a single GPU (Quadro P5000); absolute  
 582 latency will differ on deployment hardware, though relative ordering should  
 583 be preserved. A fixed 4 Hz sampling rate was used throughout—lower rates  
 584 common in low-power wearables have not been evaluated.

585 A single random seed (42) was used throughout (PyTorch deterministic  
 586 mode disabled). The narrow margin between PatchTST and Mamba ( $\Delta F1$   
 587  $= 0.005$ ) approaches the limit of reliable differentiation under a single seed.

588 The Wilcoxon test assumes independent folds, which LOSO violates (train-  
 589 ing sets share 146/147 subjects). P-values are therefore descriptive indicators  
 590 of effect consistency; raw and Bonferroni-Holm corrected thresholds are re-  
 591 ported alongside effect sizes.

592 FLOPs estimated via `thop` may not capture operator fusion, framework  
 593 optimisations, or Mamba’s custom parallel scan kernel. Inference time mea-  
 594 surements provide a complementary hardware-grounded metric.

595 Ambulatory EDA is susceptible to motion artefacts and electrode noise  
 596 [38], which controlled laboratory conditions minimise. Inter-subject variabil-  
 597 ity in electrodermal responsivity [1] may affect rankings across more hetero-

598 geneous populations; a proportion of participants may be electrodermal low-  
599 responders [? ], for whom arousal classification is inherently more difficult.  
600 Repeated-trial protocols may induce SCR habituation, introducing system-  
601 atic amplitude variation across trials that is not explicitly modelled. Ambient  
602 temperature and humidity, which influence EDA through thermoregulatory  
603 sweating [37], were not recorded; while controlled laboratory conditions likely  
604 constrained environmental variability, this remains an unmeasured confound.  
605 Robustness to signal quality degradation, demographic variability, and envi-  
606 ronmental factors has not been assessed.

607 The dataset is not publicly released due to institutional privacy con-  
608 straints; implementation code will be provided upon acceptance. Full re-  
609 producibility requires public data release or independent replication.

## 610 4. Conclusion

611 This study presented a systematic equal-budget comparison of emerg-  
612 ing architectures—spanning Transformers, state space models, and improved  
613 convolutions—for subject-independent arousal classification from Electroder-  
614 mal Activity. Eight architectures across five efficiency paradigms were evalu-  
615 ated under a strict Leave-One-Subject-Out validation framework using EDA  
616 signals from 147 participants. The key finding is that Mamba—a selective  
617 state space model—achieves near-PatchTST performance ( $F1 = 0.858$  vs.  
618  $0.863$ ) at  $3.6\times$  lower latency and with 66% fewer parameters, placing it on  
619 the Pareto frontier as the most accurate architecture within the Wearable  
620 category. By jointly evaluating classification performance and computational  
621 cost across five complementary efficiency metrics, this work provides an ex-  
622 plicit characterisation of the accuracy-efficiency trade-off for EDA classifi-  
623 cation, directly informing architecture selection under specific deployment  
624 constraints.

625 The inclusion of computational efficiency as a primary evaluation met-  
626 ric reflects a pragmatic recognition: translational value in affective comput-  
627 ing depends not only on laboratory accuracy but on deployability in real-  
628 world, resource-constrained settings. The accuracy-efficiency Pareto fron-  
629 tier directly informs architecture selection under deployment constraints—  
630 wearable, mobile, or cloud—moving beyond reporting accuracy in isolation.

631 The comparison between attention-based, state-space-based, and convolution-  
632 based paradigms further contributes to the broader methodological discussion

on the necessity of architectural complexity for physiological time-series classification. The inclusion of DLinear and ModernTCN as critical baselines provides empirical evidence on whether complex temporal modelling mechanisms yield measurable benefits for EDA signals beyond what simple linear decomposition or well-designed convolutions can achieve.

Future work will explore three directions. First, self-supervised pre-training strategies—such as masked signal modeling for physiological data [41, 42]—tailored to the most promising architectures identified here may improve performance on limited physiological datasets by leveraging unlabelled data. Second, multi-modal integration with complementary biosignals such as heart rate variability and electrocardiogram [43] may enrich the information available to lightweight classifiers without substantially increasing computational cost. Third, deployment-oriented optimisation through quantisation, pruning, and knowledge distillation [8] applied to the Pareto-optimal architectures will further narrow the gap between laboratory performance and wearable deployment, enabling real-time, on-device arousal classification for mental health monitoring, stress detection, and adaptive human-computer interaction.

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### 831 **Data Availability**

832 The dataset analysed during the current study is not publicly available due  
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834 available upon acceptance.

### 835 **Declaration of Competing Interests**

836 The authors declare that they have no known competing financial interests  
837 or personal relationships that could have appeared to influence the work  
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